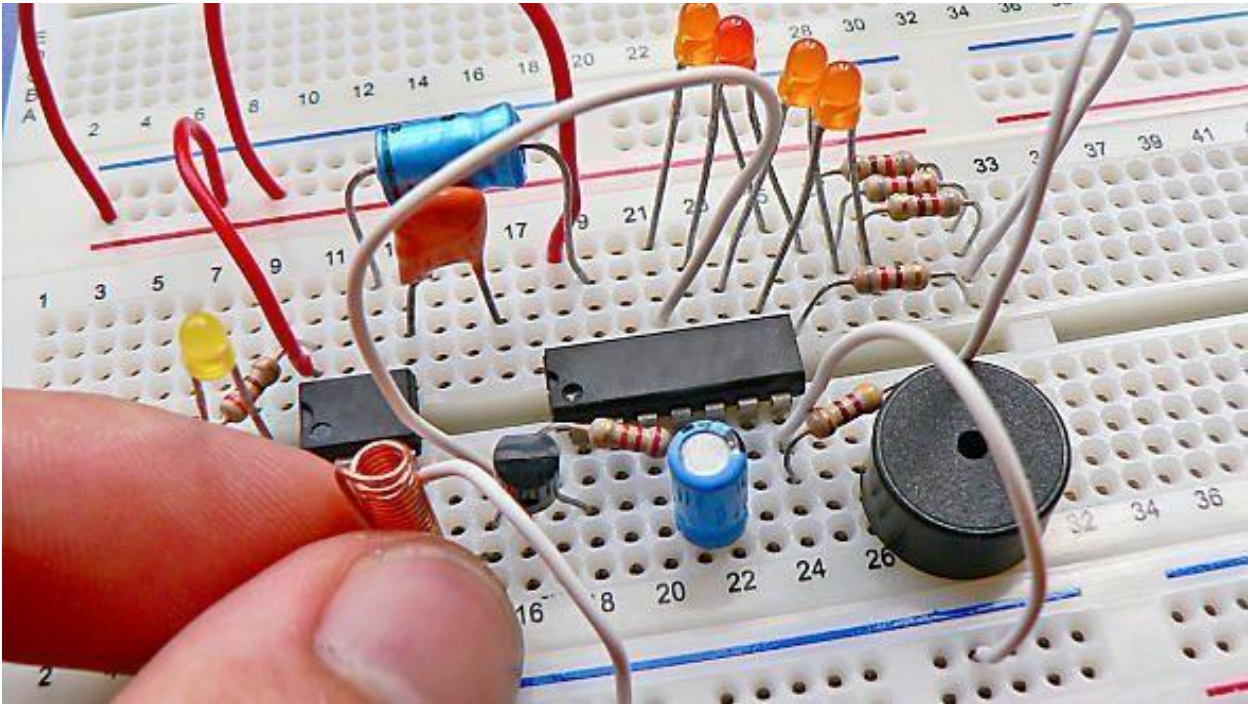


EEE109 Electronic Circuits

# Lab 2: BJT and Amplifiers



## Lab2: BJT and Amplifier Lab Script

### Lab Instructions

(1) Please read this material before the lab and perform the experiments following the instructions step by step.

(2) The instructions in **YELLOW** are the ones to be fulfilled compulsorily and included in the report.

(3) Lab 2 is composed of five sections. The first four sections are experiments to observe and analyze the behavior of BJT and amplifier with given schematics. The fifth section is the task to design a circuit to fulfill the specifications.

(4) The assessment of Lab 2 is based on the lab report,

- **Lab Report** (100 marks)

- a) This is an individual report. Each student submits his or her own lab report in PDF file via the submission link on LMO.
- b) Please use the “Lab 2 Report Template” to complete the lab report.
- c) The experimental data may be shared within and only within his or her group members. But the report must be fulfilled independently and individually.
- d) Submission deadline: available on LMO. Please be punctual. Otherwise, late submission policy may apply.

(5) **Learning Outcomes.**

A. Explain the behavior, important properties, and equivalent circuit representations, and identify various applications of diodes and transistors.

B. Create and evaluate simple rectifier and stabilizer circuits, providing detailed reasoning for chosen designs.

C. Design, construct, and critically assess simple BJT/MOSFET circuits (e.g. biasing circuit, DC analysis, AC analysis) with regards to input/output resistance, power gain, voltage gain, and current gain.

D. Perform detailed frequency response analysis of simple BJT/MOSFET amplifier circuits, considering the impact of coupling and bypass capacitors.

(6) **Artificial Intelligence Permissions.** The use of AI tools is strictly **forbidden** for this assignment.

(7) **Academic Integrity statement.** This assignment is individual work. Plagiarism (e.g., copying materials from other sources without proper acknowledgement) is a serious academic offence. Plagiarism, collusion, and academic offences involving AI will not be tolerated and will be dealt with in accordance with the University Academic Integrity Policy, potentially resulting in an overall mark of **0%** for this assessment task. Submitting work created by others, whether paid for or not, is a serious offence, and will be prosecuted vigorously. The use of AI tools beyond the scope outlined above would also be considered in breach of the University Academic Integrity Policy, and dealt with accordingly.

## **Lab2: BJT and Amplifier Lab Script**

(8) **Late submissions.** The standard University policy on late submissions will apply: **5%** of the total marks available for the assignment shall be deducted from the assessment mark for each working day after the submission date, up to a maximum of five working days, so long as this does not reduce the mark below the pass mark (50%); submissions more than five working days late will not be accepted.

## Lab2: BJT and Amplifier Lab Script

### Components and Instruments

**Platform:** NI ELVIS III

#### Section 1: Bipolar Junction Transistors

- 2N3904 transistor
- 33 k $\Omega$  resistor
- 100  $\Omega$  resistor
- 1 k $\Omega$  potentiometer on protoboard

#### Section 2: Bipolar Junction Transistor as a Switch

Calculation only

#### Section 3: Transistor Amplifiers

- 560  $\Omega$  resistor
- 1 k $\Omega$  resistor
- 8.2 k $\Omega$  resistor
- 18 k $\Omega$  resistor
- 1  $\mu$ F capacitor
- 100  $\mu$ F capacitor (Pay attention to the polarity)
- NPN 2N3904 transistor

#### Section 4: Emitter Follower Amplifiers

Simulation only

#### Section 5: Comprehensive Design Task

- (4) 1N4001 diodes
- (1) 1N4735 Zener diode
- (1) 10 k $\Omega$  potentiometer (on Protoboard)
- Proper resistors you have used in the previous sections.
- Any other passive components available in the laboratory for you to improve the DC performance.

## Section 1: Bipolar Junction Transistors

### 1.1 BJT Behavior

#### Circuit Build Instructions

1. From the Instruments tab of Measurements Live, open the following instruments:
  - IV Analyzer

**Note:** For more information about accessing Measurements Live and launching instruments, visit <http://www.ni.com/documentation/en/ni-elvis-iii/latest/gettingstarted/launching-soft-front-panels/>.

2. Place the transistor in the **Base**, **DUT+** and **DUT-** terminals in the NI ELVIS III, as indicated in the instrument's window.
  - Place the *Base* of the transistor in **BASE**.
  - Place the *Collector* of the transistor in **DUT+**.
  - Place the *Emitter* of the transistor in **DUT-**.

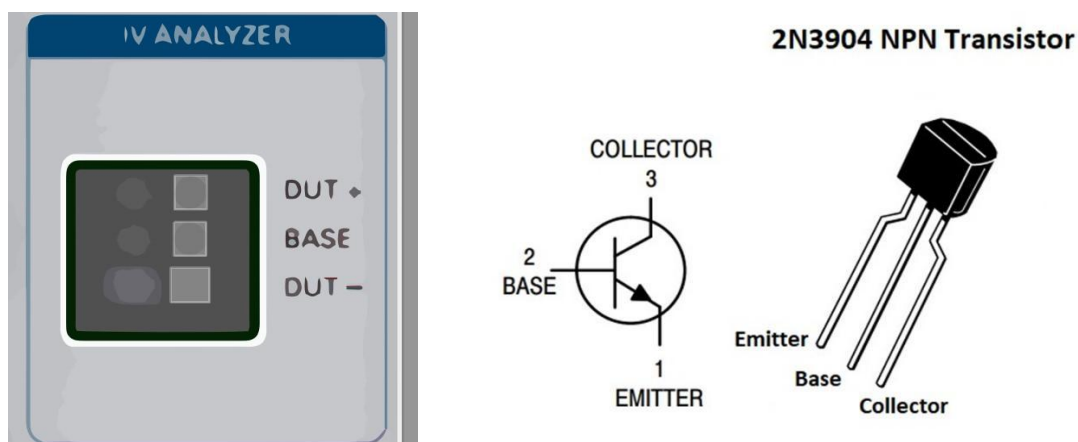


Figure 1-1 IV Analyzer and the diagram of 2N3904 transistor

3. Set the IV Analyzer according to the settings in the table below:

Table 1-1 IV Analyzer Settings

## Lab2: BJT and Amplifier Lab Script

|                                |                                 |
|--------------------------------|---------------------------------|
| Analyzer Mode                  | Transistor                      |
| Transistor Type                | NPN                             |
| Collector resistance           | Low (The first from left 左起第一格) |
| <b>Collector Voltage Sweep</b> |                                 |
| $V_c$ Start                    | 0.00 V                          |
| $V_c$ Step                     | 0.10 V                          |
| $V_c$ Stop                     | 3.00 V                          |
| $I_c$ Limit                    | 30 mA                           |
| <b>Base Current Sweep</b>      |                                 |
| $I_b$ Start                    | 15.00 $\mu$ A                   |
| $I_b$ Step                     | 11.00 $\mu$ A                   |
| Number of Curves               | 5                               |

4. The graph was generated by the IV Analyzer. Include it with your lab report.

## Lab2: BJT and Amplifier Lab Script

### 1.2 BJT Measurements in the Active Region

#### Circuit Build Instructions

**Note:** You will now obtain curves for the active region manually using the power supply and Multimeter.

1. Use the NI ELVIS III to set up the following circuit:

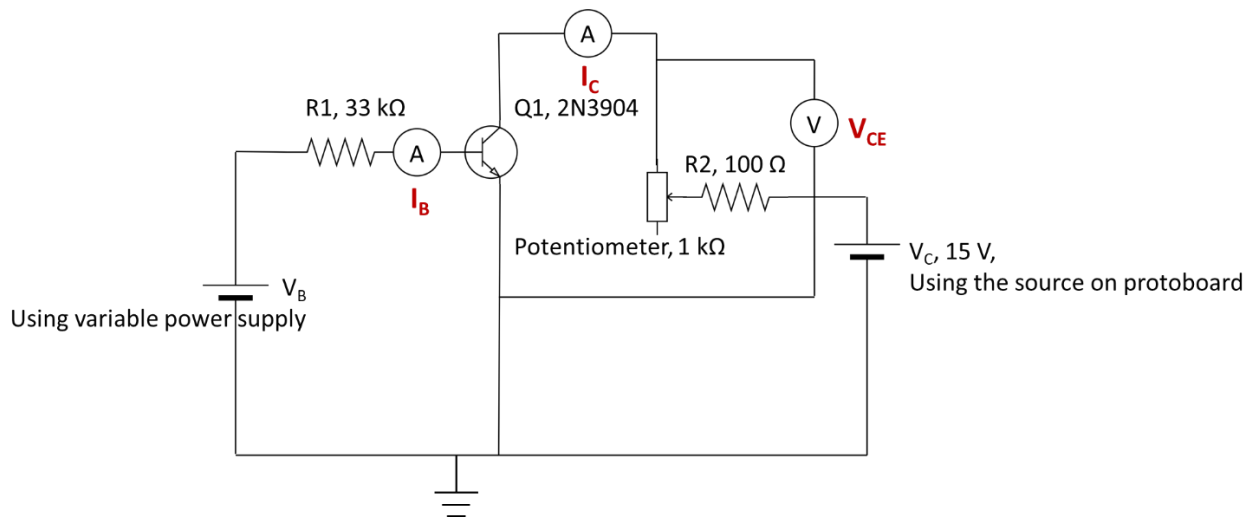


Figure 1-2 Circuit Setup

2. Open the following instruments from Measurements Live:
  - Digital Multimeter (DMM)
  - Variable Power Supply 1-15 V DC (VPS)

#### Manually Observe the Behavior of a BJT

1. In the Variable Power Supply window, apply the following settings:

Table 1-2 Variable Power Supply settings

|                           |          |
|---------------------------|----------|
| Power supply “+”          | Static   |
| Voltage (V <sub>B</sub> ) | 2.38 V   |
| Power supply “-”          | Inactive |

2. Measure the collector-emitter voltage (V<sub>CE</sub>) using the DMM measuring DC voltage. Manually adjust the potentiometer until the reading is 6 V. If the V<sub>CE</sub> cannot be adjusted to 6 V, you can start it from a higher voltage, such as 6.5 V or 7 V.

## Lab2: BJT and Amplifier Lab Script

3. Read the collector-emitter voltage ( $V_{CE}$ ), as well as the collector's current ( $I_C$ ), and write the data in Table 1-3.
4. Change the voltage  $V_C$  using the data in the table below, repeat the  $V_{CE}$  and  $I_C$  readings, and write the data in the table.
5. Repeat for voltage ( $V_B$ ) of 4.08 V and 5.76 V, until you complete the table below:

Table 1-3

|                                                            |                   |                                                             |                   |                                                             |                   |
|------------------------------------------------------------|-------------------|-------------------------------------------------------------|-------------------|-------------------------------------------------------------|-------------------|
| $V_B = 2.38\text{ V}$<br>( $I_B = 50\text{ }\mu\text{A}$ ) |                   | $V_B = 4.08\text{ V}$<br>( $I_B = 100\text{ }\mu\text{A}$ ) |                   | $V_B = 5.76\text{ V}$<br>( $I_B = 150\text{ }\mu\text{A}$ ) |                   |
| $V_{CE}\text{ (V)}$                                        | $I_C\text{ (mA)}$ | $V_{CE}\text{ (V)}$                                         | $I_C\text{ (mA)}$ | $V_{CE}\text{ (V)}$                                         | $I_C\text{ (mA)}$ |
| 6                                                          |                   | 6                                                           |                   | 6                                                           |                   |
| 7                                                          |                   | 7                                                           |                   | 7                                                           |                   |
| 8                                                          |                   | 8                                                           |                   | 8                                                           |                   |
| 9                                                          |                   | 9                                                           |                   | 9                                                           |                   |
| 10                                                         |                   | 10                                                          |                   | 10                                                          |                   |
| 11                                                         |                   | 11                                                          |                   | 11                                                          |                   |
| 12                                                         |                   | 12                                                          |                   | 12                                                          |                   |

### Generate a Collector Current vs. Collector-Emitter Voltage Graph

1. Using the data from the Table 1-3 in the previous step, generate a Collector-Emitter Voltage (x) vs. Collector Current (y) graph. **Include it with your lab report.**

You have two options:

- You can create the graph by clicking on the Excel file below and submit a screenshot or photograph of your graph.

OR

- You can manually draw the graph and submit a photo of the graph with your lab.

**Questions 1-1** Is  $\beta$  relatively constant on all points?

A. Yes

B. No

If not, what is different?

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## Lab2: BJT and Amplifier Lab Script

**Questions 1-2** What is the transistor's approximate  $\beta$ ? **Tip:** Use a base current  $I_B = 100 \mu\text{A}$  at  $V_{CE} = 8 \text{ V}$ .

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**Questions 1-3** What would be the collector's  $I_C$  current if  $V_{CE} = 9 \text{ V}$  at  $I_B = 75 \mu\text{A}$ ?

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### 1.3 Conclusion

These questions will help you review and interpret the concepts learned in this section of the lab.

**Questions 1-4** Use the chart below to compare the procedures from the simulation, the IV Analyzer and the instrumentation measurement activities. What are their advantages and disadvantages?

Table 1-4

|                 | Advantages | Disadvantages |
|-----------------|------------|---------------|
| Simulation      |            |               |
| IV Analyzer     |            |               |
| Instrumentation |            |               |

## Section 2: Bipolar Junction Transistor as a Switch

### 2.1 BJT as a Switch

#### Circuit Build Instructions

1. For the illustration below, you will need to calculate the following:
  - Cutoff collector-emitter voltage  $V_{CE}$
  - Saturation collector-emitter voltage  $V_{CE}$
  - Saturation voltage across the collector resistor  $V_{RC}$
  - Saturation current at collector  $I_C$
2. Complete Table 2-1 with the values you calculated.

**Note:** A few things to remember for this question:

- The LED voltage for cutoff and saturation purposes is 1.8 V.
- During the saturation stage, the transistor will be in full operation.
- $V_{CE}$  is 0.1 V (typical value).

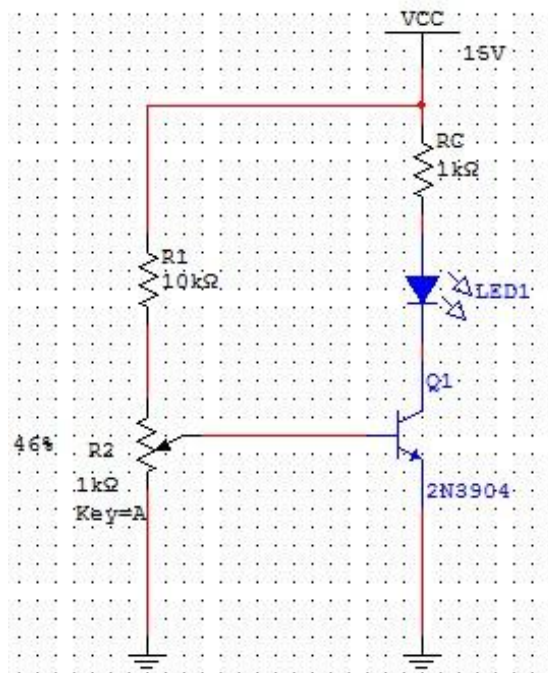


Figure 2-1

3. Record your calculated values in the table below:

Calculate  $V_{RC}$  by applying Kirchhoff's voltage law to the transistor's collector.

## Lab2: BJT and Amplifier Lab Script

Calculate saturation  $I_C$  using Ohm's law in  $V_{RC}$ .

Table 2-1

| Data                  | Calculated Values |
|-----------------------|-------------------|
| $V_{CE}$ (cut-off)    |                   |
| $V_{CE}$ (saturation) |                   |
| $V_{RC}$ (saturation) |                   |
| $I_C$ (saturation)    |                   |

## 2.2 Conclusion

These questions will help you review and interpret the concepts learned in the lab.

**Questions 2-1** List the most important characteristics and applications of bipolar junction transistors.

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**Questions 2-2** Explain the different regions of operation of a bipolar junction transistor.

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## Section 3: Transistor Amplifiers

### 3.1 Transistor Amplifiers

#### Circuit Build Instructions

You will now expand what you know about transistor amplifiers by completing the following activity.

1. Use the NI ELVIS III to set up the circuit shown in the following illustration:

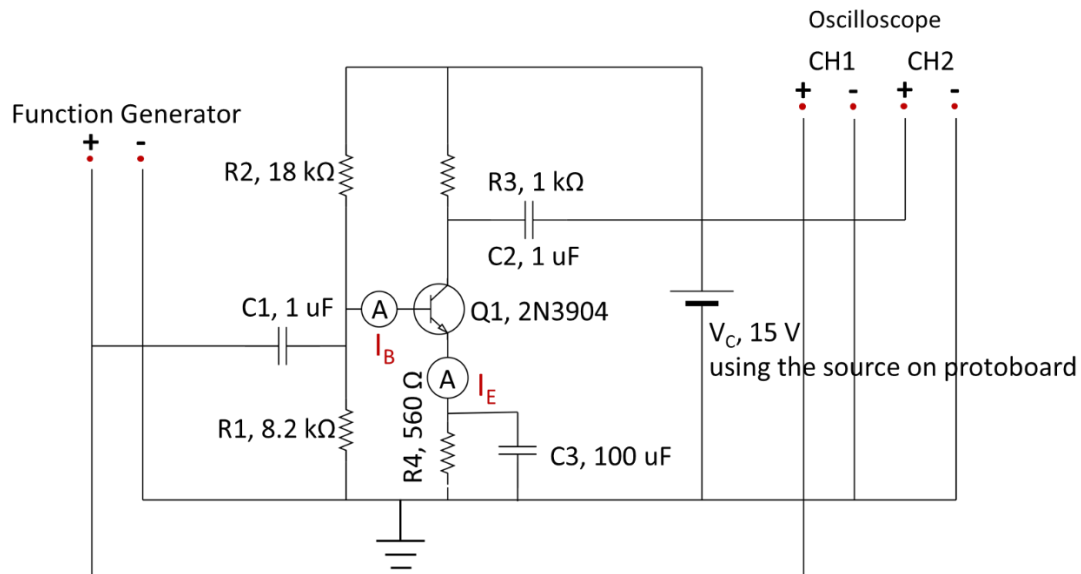


Figure 3-1 Circuit Setup

Pay attention to the polarity of the capacitor (注意电容的极性，长进短出)

2. From the Instruments tab of Measurements Live, open the following instruments:
  - Digital Multimeter (DMM)
  - Function Generator (FGen-Arb)
  - Oscilloscope

**Note:** For more information about accessing Measurements Live and launching instruments, visit <http://www.ni.com/documentation/en/ni-elvis-iii/latest/gettingstarted/launching-soft-front-panels/>.

3. Start the Function Generator with the following settings:

Table 3-1 Function Generator settings

|           |         |
|-----------|---------|
| Frequency | 100 Hz  |
| Amplitude | 0.2 Vpp |

## Lab2: BJT and Amplifier Lab Script

Given a  $\beta$  of 100 and  $V_{BE}$  of 0.7, calculate  $I_B$  and  $I_E$ , and write the calculated results in Table 3-2 below.

After you have calculated your results, use the DMM to measure  $I_B$  and measure  $I_E$ . You can measure from the exposed metal leads or create a measurement point using a wire or header pin. Write the measured values in Table 3-2 below.

Table 3-2

| Data  | Calculated Value | Measured Value |
|-------|------------------|----------------|
| $I_B$ |                  |                |
| $I_E$ |                  |                |

Table 3-3

| Data     | Given Value | Measured/Calculated Value |
|----------|-------------|---------------------------|
| $\beta$  | 100         |                           |
| $V_{BE}$ | 0.7         |                           |

### Determine Gain Using the Oscilloscope

1. Using your oscilloscope measure:
  - The function generator's signal at one channel
  - The amplifier's output on the other channel
2. Adjust the Time/Div and Volts per division to see both signals at once.

**Include the figure in your lab report.**

3. Calculate linear gain (Output  $V_{pp}$ / Input  $V_{pp}$ ). Calculate gain in dB and write the results below.  
Oscilloscope Gain (dB) =

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## 3.2 Frequency Response of the Amplifier

1. Click Stop on the Function Generator and the Oscilloscope.

## Lab2: BJT and Amplifier Lab Script

2. Open the Bode Analyzer.
3. Set the Bode Analyzer as indicated below:

*Table 3-3 Bode Analyzer Settings*

| Stimulus Channel |       |
|------------------|-------|
| Start Frequency  | 10 Hz |
| Stop Frequency   | 5 kHz |
| Cursors          | On    |

4. Perform the reading and analyze the results by pressing **Run**.

Use the cursors to find the gain at 100 Hz and write it down. **Mention this result in your lab report.**

Bode Gain (dB) =

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Does the measured gain coincide with the gain found using the oscilloscope? **Mention this result in your lab report.**

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### 3.3 Conclusion

These questions will help you review and interpret the concepts learned in this section of the lab.

**Questions 3-1** List the most important characteristics and applications of a transistor amplifier.

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**Questions 3-2** Discuss how, in the previous steps, you measured the voltage gain of a transistor amplifier circuit.

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## Lab2: BJT and Amplifier Lab Script

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**Questions 3-3** What happens to the amplified signal from a transistor amplifier if we increase the input signal amplitude?

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## Section 4: Emitter Follower Amplifiers

### 4.1 Simulate: Emitter Follower Amplifiers – Voltage Response

As learned in Section 3, the main role of a transistor is to take a relatively small input signal and make it larger by increasing the signal voltage. You also looked at the three standard BJT amplifier configurations.

In this section, you will build a configuration of components that form a commonly-used transistor amplifier called the emitter follower or common collector. With this amplifier, the voltage gain is always slightly less than 1. This means the voltage will not increase but the current gain can be higher than 1. Emitter followers are commonly used to amplify the current of a signal, usually for a low impedance load such as a speaker.

In this activity, you will investigate the voltage response of an emitter follower circuit using Multisim Live.

#### Circuit Build Instructions

1. Click the link below to open the Multisim Live circuit.

<https://www.multisim.com/content/zGWUhyntWEfCHexu6GREc/emitterfollower-v/open>

#### Observe the Voltage Response

1. Examine the emitter follower circuit, simulated with Multisim Live.

**Note:** The AC signal source produces a 10 kHz, 5 Vpp (2.5 V Pk) signal.

2. Click Run.
3. Measure the voltage response using the probes PR1 and PR2.
  - PR1 and PR2 should display essentially the same waveform. Hover your cursor over different points on the output curve and observe how the cursor shifts between the blue input plot and the green output plot.

**Note:** The output voltage has almost exactly the same phase and amplitude as the input voltage. This is the reason that this amplifier is called a follower. The voltage output follows the voltage wired to the base of the transistor.

#### Observe the Current Response

Now observe the current gain in an emitter follower amplifier.

1. Click the link below to open the Multisim Live circuit

<https://www.multisim.com/content/B2DuX5YtreVZMtNVj6r2gW/emitter-followera/open>

#### Record Peak-to-Peak Current Values

1. Click Run.



## Lab2: BJT and Amplifier Lab Script

2. Measure the current response using the probes PR1 and PR2.

Use the table below to record the peak-to-peak current from Probe 1 and Probe 2.

Table 4-1

| Probe                    | Peak-to-Peak Current (μA) |
|--------------------------|---------------------------|
| Probe 1 – Input Current  |                           |
| Probe 2 – Output Current |                           |

Calculate the current gain using the following equation:

$$A_I = \frac{I_{out}}{I_{in}} =$$

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3. Click Stop when you are done. **Mention this result in your lab report.**

### 4.2 Simulate: Identify the Relationship between Resistors and Current Gain

Typically, the load of a circuit ( $R_L$ , in this simulation) is not going to change. If we assume that the load in a real-life example of this circuit is an audio speaker, the speaker is not going to change its physical properties.

It would be more appropriate to change the resistance  $R_E$  to examine how the current gain will respond.

1. Modify the resistance of  $R_E$  by clicking its value and typing or using the slider to select a new one.
2. Calculate the circuit gain for a range of resistance values by:
  - Modifying the resistance ( $R_E$ )
  - Running the simulation
  - Making the measurement
  - Stopping the simulation

Then, complete Table 4-2 below:

## Lab2: BJT and Amplifier Lab Script

Table 4-2

| Resistance of RE | Probe 1 – Input Current ( $\mu\text{A}$ ) | Probe 2 – Output Current ( $\mu\text{A}$ ) | Gain |
|------------------|-------------------------------------------|--------------------------------------------|------|
| 1k               |                                           |                                            |      |
| 2k               |                                           |                                            |      |
| 3k               |                                           |                                            |      |
| 5k               |                                           |                                            |      |

Mention this result in your lab report.

**Note:** The relationship between RE and gain is non-linear. Because the load is going to be different for every application, the combination of resistors in a transistor amplifier is often customized for a specific application.

### 4.3 Conclusion

These questions will help you review and interpret the concepts learned in the lab.

**Questions 4-1** In your own words, talk about the purpose of an emitter follower amplifier.

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**Questions 4-2** Give a few application examples of a transistor amplifier.

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## Lab2: BJT and Amplifier Lab Script

### Section 5: Comprehensive Design Task

You are supplied with the following components and instructions. Please design a signal detection circuit to drive the LED for indication. When the given frequency of sine wave signal is applied to the input of the designed circuit, the LED is ON. When the signal disappears, the LED is OFF.

#### Input:

- 15V DC as power supply
- Sine wave generated by the Function Generator with the following settings:

*Table 5-1 Function Generator settings*

|           |                     |
|-----------|---------------------|
| Frequency | 100 Hz              |
| Amplitude | 0.2 V <sub>pp</sub> |
| DC Offset | 0 V                 |

#### Output:

- LED will be ON or OFF depending on whether the sine wave signal is applied or not.

#### Components:

1. LED indicator as output
2. 2N3904 transistors
3. Resistors provided by the laboratory
4. Capacitors provided by the laboratory
5. (Optional) Diodes available from LAB 1.
6. Any other passive components available in the laboratory

#### Instruments:

1. Function Generator (FGen-Arb)
2. Digital Multimeter (DMM)
3. Oscilloscope

#### Hints for Design:

The following circuit modules may be recommended for the design,

1. The sine wave signal may be amplified for driving the following stage.

## Lab2: BJT and Amplifier Lab Script

2. The amplified signal may be AC coupled to the second stage for it to be rectified to a DC signal, either in voltage or in current. Please refer to the knowledge in LAB 1. The rectification may be implemented by diodes (LAB 1) or source follower (LAB 2). The source follower has BE junction to perform the active rectification with better driving capability due to the current gain.
3. The DC signal may be DC coupled to the Switch circuit of BJT, which is utilized to drive LED to ON or OFF. The output of rectifier may be tuned to match the Switch stage so that it works between saturation and cut-off regions.
4. Build the circuit in individual stages before cascading them together. The stage by stage debug tricks is recommended. Say, larger sine wave signal may be used as input to tune rectification stage. The appropriate DC signal levels may be used as input to Switch stage to check if it works or not. Similar tricks apply to amplifying stage as well.
5. When debugging, always make sure VDD is correctly connected and the critical biasing of each stage matches what you designed.

### Instructions

1. Draw the designed schematic with detailed explanation of the function of each stage and their components. **Include them in the report.**
2. Build the circuit on NI ELVIS III platform.

**Take photos of the breadboard circuits. Insert it in the lab report. Group number label on the NI ELVIS III platform MUST be clearly shown in the photos.**

3. Setup the Measurements Live so that the Function Generator, Oscilloscope and Multimeter are shown in the same screen. Measure the input by Oscilloscope to show the sine wave signal. Measure the output nodes connecting LED simultaneously by Multimeter to show the DC levels when ON or OFF.
4. Take two screenshots. One is that when the given sine wave is applied to the input, the Multimeter reads an output level so LED is ON. The other is when the given sine wave is stopped, the Multimeter reads another output level so LED is OFF. **Include the two in the report.**